RELATIONAL DATABASE MANAGEMENT SYSTEM

HOME ASSIGNMENT

Module I: Introduction to DBMS

1. Define a DBMS and explain its objectives, major functions, and applications. Compare a file-processing system with a DBMS.
2. Draw and explain the architecture of a DBMS and describe its major components.
3. Compare the network, hierarchical, and relational data models with suitable examples.
4. Explain the roles and responsibilities of different database users. Differentiate database administrator, database designer, application programmer, and end user.
5. Explain database languages, including DDL, DML, DCL, and transaction-control commands, with examples.
6. Differentiate schema and instance. Explain data independence and distinguish logical data independence from physical data independence.

Module II: Data Modeling and Relational Model

7. Design an ER diagram for a college database containing Students, Courses, Faculty, and Enrolments. State the entities, attributes, keys, relationships, and cardinalities.
8. Explain the Enhanced ER (EER) model. Discuss specialization, generalization, and their constraints with examples.
9. Define the relational database model and explain its properties, tuples, attributes, domains, degree, and cardinality.
10. Explain primary, candidate, super, alternate, and foreign keys. Discuss entity integrity, referential integrity, and other relational constraints.
11. Explain relational algebra operations (selection, projection, union, set difference, Cartesian product, rename, and joins) using example relations. Also explain relational calculus and distinguish tuple relational calculus from domain relational calculus.
12. ***Draw ER diagram for the given scenario:***
13. Consider a bus ticketing system that records information about the **passegner, bus** and **route**. Passenger is assigned to a bus that travels to a route. A bus contains many passegners and a passegner can be assigned into only one bus. Many busses travel in same route but a bus can travel in only one route. The attributes of passegner are pid(unique), gender and phone_no(multi-values). Similarly bus contains regno(unique) and color and route contains rid(unique), distance and rate.
14. Let
    R = {A, B, C, D, E}
    and

    F = { AB $\longrightarrow$ C, A $\longrightarrow$ D, D $\longrightarrow$ E, AC $\longrightarrow$ B }

    Then compute $\{AB\}^+$ and $\{DC\}^+$.
15. Consider the relation:

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student(sid, sname, address, gender, age)
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Write relational algebra(RA) for the following:

To display name of student whose age is greater than 15.
To remove the record of the student who are from Kathmandu.
To update address of student to Kathmandu whose sid is 'S1101'.

16. ***Consider the following relational database:***
    Employee(empid, gender, name, dob, salary, address, deptno, start_date)
    Department(dpetno, name)
    Students(std_id, gender, name, dob, projectno, deptno)
    Project(projectno, name, location)

    Write **Relational algebra** expression for the following:
    Find employee name and address who works in "xyz company".
    Delete all employees whose salary is greater than Rs. 1,00,000.
    Update department name to "SitaGroup" where student "Sita" is involved.
    Find the average salary of all employees.
    Insert student record named "Gopal Shrestha" for department "Ram Group".
17. Design a database using ER diagram for a commercial bank. This commercial bank maintains information about entities: Depositor, Account, Loan, and Branch. State any assumptions made in the design of the ER diagram.
18. Consider the following relation tor published books:

    BOOK (Book title, Author_ name, Book_type, List price, Author affl, Publisher)
    Author affil refers to the affiliation of author. Suppose the following dependencies exist :
    Book_title → Publisher, Book_type
    Book_type → List_price
    Author_name → Author_affil
    a. What normal form is the relation in? Explain your answer.
    b. Apply normalization until you cannot decompose the relations further.
    State the reasons behind each decomposition.

19. Given the SQL query below, show all steps of query optimization — converting it into relational algebra, constructing the initial query tree, and applying heuristic transformations to obtain an optimized query tree:
    SELECT Pname FROM PROJECT WHERE Plocation = 'Houston' AND Pnumber > 10;

Module III: Relational Database Design

20. Define functional dependency. Explain trivial, non-trivial, full, partial, and transitive functional dependencies with examples.
21. Explain normalization and the problems caused by insertion, deletion, and update anomalies.
22. Normalize a relation of your choice step by step into 1NF, 2NF, and 3NF. Clearly identify the functional dependencies at each stage.
23. Explain Boyce–Codd Normal Form (BCNF) and compare it with 3NF. Give an example where a relation satisfies 3NF but not BCNF.

Module IV: Query Language — SQL and PL/SQL

24. Explain SQL components: DDL, DML, and DCL. Give at least two commands and examples for each component.

25. Using suitable tables, write SQL queries demonstrating SELECT, FROM, WHERE, ORDER BY, GROUP BY, and HAVING. Explain the difference between WHERE and HAVING.
26. Describe the structure of a PL/SQL block. Explain declarations, executable statements, and exception-handling sections with an example.
27. Explain PL/SQL control statements, including conditional and iterative statements, with examples.
28. Explain implicit and explicit cursors and demonstrate cursor processing in PL/SQL.
29. Differentiate procedures and functions. Write a simple example of each.
30. Explain database triggers, their types, and their applications. Write a trigger example that checks a condition before an INSERT or UPDATE.

Module V: Database Security and Authorization

31. Discuss common database security issues and threats. Explain the importance of confidentiality, integrity, and availability.
32. Explain Discretionary Access Control (DAC) and Mandatory Access Control (MAC), and compare their principles.
33. Explain user privileges, roles, GRANT, and REVOKE commands with SQL examples.
34. What is statistical database security? Explain inference-related risks and methods used to protect statistical information.

Module VI: Transaction Management and Concurrency Control

35. Define a transaction and explain the ACID properties with a real-world example.
36. Explain schedules and distinguish serial and non-serial schedules. Define recoverable, cascadeless, and strict schedules.
37. Explain conflict serializability and view serializability. Demonstrate how a precedence graph is used to test conflict serializability.
38. Describe common concurrency problems: lost update, dirty read, unrepeatable read, and incorrect summary, using examples.
39. Explain locking protocols, including binary locks, shared/exclusive locks, and two-phase locking (2PL).
40. Explain the timestamp-ordering protocol and multiversion concurrency-control technique. Compare their advantages and limitations.
41. Define deadlock. Explain deadlock prevention, detection, and resolution, including the wait-for graph.

Module VII: Database Backup and Recovery

42. Explain database failures and the need for backup and recovery. Distinguish transaction, system, and media failures.
43. Compare immediate-update and deferred-update recovery techniques.
44. Explain Write-Ahead Logging (WAL) and the role of log records during recovery.
45. Describe the ARIES recovery algorithm and explain its analysis, redo, and undo phases.
46. Explain shadow paging, including its working, advantages, and limitations. Compare shadow paging with log-based recovery.